

A regular infinite-dimensional Pick generator with real continuous spectrum

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Abstract

Schober asks whether the finite-dimensional criteria for regular operator-valued Pick functions extend to infinite-dimensional Hilbert spaces. They do not extend verbatim. On $\mathcal{H} = \ell^2(\mathbb{N})$ let $De_n = n^{-1}e_n$ and $F(z) = zD$. The associated boundary group is the direct sum of rescaled standard shifts. An explicit coordinatewise dilation conjugates it to the standard shift of multiplicity ℓ^2 , so F is regular and its multiplicity space is ℓ^2 . Nevertheless

$$0 \in \text{Spec}(F(z)) \quad (z \in \mathbb{C}_+), \quad D(F, 0) = \mathbb{R}.$$

Thus conditions (b), (d), and (e) of source Theorem 3.1.7 and the zero-set condition of Theorem 3.1.10 fail for a regular infinite-dimensional Pick function. Condition (c), involving actual kernels, still holds; the precise obstruction is continuous spectrum accumulating at zero.

1 Source problem

The source is J. Schober, *Regular one-parameter groups, reflection positivity and their application to Hankel operators and standard subspaces*, arXiv:2406.04241v2 [1]. Its introduction highlights Theorems 3.1.7 and 3.1.10, which characterize regular Pick generators in finite dimension, and asks whether similar results hold in infinite dimension.

via some isomorphism of regular one-parameter groups (cf. Theorem 2.3.9).

In Chapter 3, we study the semigroup $\text{Inn}(\mathbb{C}_+, B(\mathcal{H}))$ of inner functions appearing in the context of the Longo–Witten Theorem. One-parameter semigroups in $\text{Inn}(\mathbb{C}_+, B(\mathcal{H}))$ are infinitesimally generated by operator-valued Pick functions (cf. Definition A.4.1), i.e. they are of the form

$$U_t^F := e^{itF}, \quad t \in \mathbb{R},$$

for some Pick function F with values in $B(\mathcal{H})$. For $\dim \mathcal{H} < \infty$ we give criteria on the function F that are equivalent to $(L^2(\mathbb{R}, \mathcal{H}), H^2(\mathbb{C}_+, \mathcal{H}), U^F)$ being a regular one-parameter group (cf. Theorem 3.1.7, Theorem 3.1.10). In these cases, we are also interested in calculating the multiplicity space $\mathcal{K}_F$ of the regular one-parameter group $(L^2(\mathbb{R}, \mathcal{H}), H^2(\mathbb{C}_+, \mathcal{H}), U^F)$. A formula for this multiplicity space $\mathcal{K}_F$ is provided in Theorem 3.3.27. An interesting open problem in this context is whether similar results also hold in the case of an infinite-dimensional Hilbert space $\mathcal{H}$.

The principal finite-dimensional criterion is:

Theorem 3.1.7. *Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space and $F \in \text{Pick}(\mathbb{C}_+, B(\mathcal{H}))_{\mathbb{R}}$. Then the following are equivalent:*

- (a) F is regular.
- (b) $\text{Spec}(F(z)) \subseteq \mathbb{C}_+$ for all $z \in \mathbb{C}_+$.
- (c) $\ker(x\mathbf{1} - M_{F_*}) \cap H^2(\mathbb{C}_+, \mathcal{H}) = \{0\}$ for all $x \in \mathbb{R}$.
- (d) $\mathbb{R} \setminus D(F, x)$ has non-zero Lebesgue measure for all $x \in \mathbb{R}$.
- (e) $D(F, x)$ is a Lebesgue zero-set for all $x \in \mathbb{R}$.

Proof. (a) $\Rightarrow$ (c): Let F be regular and $x \in \mathbb{R}$. Further, let $v \in \ker(x\mathbf{1} - M_{F_*}) \cap H^2(\mathbb{C}_+, \mathcal{H})$. Then

$$\{0\} = \bigcap_{t \in \mathbb{R}} e^{itF_*} H^2(\mathbb{C}_+, \mathcal{H}) \supseteq \bigcap_{t \in \mathbb{R}} e^{itF_*} \mathbb{C}v = \bigcap_{t \in \mathbb{R}} e^{itx} \mathbb{C}v = \mathbb{C}v,$$

The second criterion says that regularity is equivalent to preservation of Lebesgue zero-sets under $M \mapsto D(F, M)$:

Proof. By Theorem 3.1.7 for $z \in \mathbb{C}_+$, we have

$$\{f(z)\} = \text{Spec}(f(z)) \subseteq \mathbb{C}_+, \quad \{g(z)\} = \text{Spec}(g(z)) \subseteq \mathbb{C}_+ \quad \text{and} \quad \text{Spec}(F(z)) \subseteq \mathbb{C}_+.$$

Therefore, for $z \in \mathbb{C}_+$, we get

$$\text{Spec}((f \circ F \circ g)(z)) = f(\text{Spec}(F(g(z)))) \subseteq f(\mathbb{C}_+) \subseteq \mathbb{C}_+,$$

which by Theorem 3.1.7 implies that $f \circ F \circ g$ is regular. $\square$

With help of this corollary, we can prove a theorem with a generalization of condition (e) in Theorem 3.1.7

Theorem 3.1.10. *Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space and $F \in \text{Pick}(\mathbb{C}_+, B(\mathcal{H}))_{\mathbb{R}}$. Then the following are equivalent:*

- (a) F is regular.
- (b) $D(F, M)$ is a Lebesgue zero-set for all Lebesgue zero-sets $M \subseteq \mathbb{R}$.

We use the natural unchanged definitions for an infinite-dimensional complex Hilbert space $\mathcal{H}$. Thus a norm-holomorphic $F : \mathbb{C}_+ \rightarrow B(\mathcal{H})$ is Pick if $\text{Im } F(z) \geq 0$, it has self-adjoint boundary values $F_*(x)$ almost everywhere, and it is regular if

$$\bigcap_{t \in \mathbb{R}} e^{itF_*} H^2(\mathbb{C}_+, \mathcal{H}) = \{0\}, \quad \overline{\bigcup_{t \in \mathbb{R}} e^{itF_*} H^2(\mathbb{C}_+, \mathcal{H})} = L^2(\mathbb{R}, \mathcal{H}).$$

2 Proof intuition

Finite dimension does not distinguish spectrum from approximate point spectrum in the way needed here: a spectral point on the real axis is an eigenvalue and produces a fixed Hardy-space direction. In infinite dimension, strictly positive eigenvalues may accumulate at zero. Taking the orthogonal sum of the scalar regular functions z/n preserves regularity, but zero enters the spectrum without creating any zero eigenvector. This breaks every condition in the source that tests the full spectrum while leaving the kernel condition intact.

3 A diagonal family

The counterexample is a special case of a transparent family.

Lemma 1. *Let $(a_n)_{n \geq 1}$ be a bounded sequence of strictly positive real numbers, let $\mathcal{H} = \ell^2(\mathbb{N})$, and set*

$$De_n = a_n e_n, \quad F(z) = zD \quad (z \in \mathbb{C}_+).$$

Then F is an operator-valued Pick function with self-adjoint boundary values and is regular. More precisely,

$$(L^2(\mathbb{R}, \mathcal{H}), H^2(\mathbb{C}_+, \mathcal{H}), U^F) \cong (L^2(\mathbb{R}, \ell^2), H^2(\mathbb{C}_+, \ell^2), S),$$

so its multiplicity space is ℓ^2 .

Proof. The function F is norm-holomorphic and

$$\operatorname{Im} F(z) = (\operatorname{Im} z)D \geq 0.$$

Its boundary values are $F_*(x) = xD$, which are self-adjoint. Hence the associated group acts coordinatewise by

$$(U_t^F f)_n(x) = e^{ita_n x} f_n(x).$$

For each n define a positive-dilation unitary on $L^2(\mathbb{R})$ by

$$(W_n f)(y) = a_n^{-1/2} f(y/a_n).$$

Positive dilation preserves $H^2(\mathbb{C}_+)$, and a direct calculation gives

$$W_n M_{e^{ita_n x}} = M_{e^{ity}} W_n.$$

Therefore $W = \bigoplus_{n \geq 1} W_n$ is a unitary on $L^2(\mathbb{R}, \ell^2)$, maps $H^2(\mathbb{C}_+, \ell^2)$ onto itself, and satisfies

$$W U_t^F W^{-1} = S_t = M_{e^{ity}} \otimes \mathbf{1}_{\ell^2} \quad (t \in \mathbb{R}).$$

The standard shift triple on the right is a regular one-parameter group with multiplicity space ℓ^2 . Hence so is the triple generated by F . $\square$

4 Counterexample to both finite-dimensional criteria

Theorem 2. *On $\mathcal{H} = \ell^2(\mathbb{N})$ let*

$$De_n = \frac{1}{n} e_n, \quad F(z) = zD.$$

Then F is regular, but:

1. $\operatorname{Spec}(F(z)) \not\subseteq \mathbb{C}_+$ for every $z \in \mathbb{C}_+$;
2. $D(F, 0) = \mathbb{R}$, so the singleton and general zero-set criteria of source Theorems 3.1.7 and 3.1.10 fail;
3. nevertheless,

$$\ker(\lambda \mathbf{1} - M_{F_*}) = \{0\} \quad \text{on } L^2(\mathbb{R}, \ell^2) \quad (\lambda \in \mathbb{R}).$$

Thus both equivalences fail in infinite dimension.

Proof. Regularity and the multiplicity statement follow from Lemma 1. The operator D is compact, positive, and injective, with

$$\text{Spec}(D) = \{0\} \cup \left\{ \frac{1}{n} : n \geq 1 \right\}.$$

For $z \in \mathbb{C}_+$ the scalar z is nonzero, so

$$\text{Spec}(F(z)) = z \text{Spec}(D) = \{0\} \cup \left\{ \frac{z}{n} : n \geq 1 \right\}.$$

Because $0 \notin \mathbb{C}_+$, condition (b) of source Theorem 3.1.7 fails.

For every $y \in \mathbb{R}$, zero belongs to $\text{Spec}(yD)$. Recalling the source notation

$$D(F, M) = \{y \in \mathbb{R} : M \cap \text{Spec}(F_*(y)) \neq \emptyset\},$$

we therefore have

$$D(F, 0) = D(F, \{0\}) = \mathbb{R}.$$

Thus $\mathbb{R} \setminus D(F, 0)$ has zero measure and $D(F, 0)$ is not a Lebesgue zero-set. This disproves conditions (d) and (e) of Theorem 3.1.7. Since $\{0\}$ itself is a Lebesgue zero-set, it also disproves the implication from regularity to the zero-set condition in Theorem 3.1.10.

Finally, suppose $(\lambda \mathbf{1} - M_{F_*})f = 0$ for some $f = (f_n)_{n \geq 1} \in L^2(\mathbb{R}, \ell^2)$. The n th coordinate gives

$$(\lambda - y/n)f_n(y) = 0 \quad \text{for almost every } y \in \mathbb{R}.$$

Outside the singleton $\{n\lambda\}$ the scalar coefficient is nonzero. Hence $f_n = 0$ as an L^2 function for every n , and therefore $f = 0$. This proves the kernel assertion and shows that the failed spectral point is continuous rather than point spectrum. $\square$

5 Verification, limitations, and novelty

The dilation formula gives an explicit isomorphism of regular one-parameter groups, so regularity does not rest on an informal direct-sum argument. For the chosen $a_n = 1/n$ it reads

$$(W_n f)(y) = \sqrt{n} f(ny),$$

which is readily checked to be unitary and to preserve the Hardy half-plane. The spectrum computation uses only the standard spectrum of a compact diagonal operator. The multiplication-kernel calculation is stronger than the source's condition (c), because it proves triviality on all of L^2 , not only on its Hardy subspace.

The source's phrase "similar results" is broad. This packet gives a full negative answer to the verbatim extension of the two explicitly cited classification theorems. It does not preclude corrected infinite-dimensional criteria formulated with point spectrum or spectral measures. Nor does it contradict a possible infinite-cardinal version of the multiplicity formula: the example has explicitly computed multiplicity ℓ^2 .

A bounded novelty search on August 13, 2026 checked all four lightweight run indexes for the arXiv id, exact title, regular operator-valued Pick functions, the spectrum condition, and the infinite-dimensional question. External searches used the exact title/id and the phrases "infinite-dimensional Hilbert", "regular Pick function", and " $\text{Spec}(F(z))$ ". They found arXiv v2 and copies of the thesis, but no later paper stating this counterexample. Priority remains subject to specialist review.

This packet is recommended for expert review as a likely valid full scoped counterexample. The main audit points are the natural infinite-dimensional extension of the source definitions, preservation of vector-valued Hardy space by the coordinatewise dilation W , and the distinction between point and continuous spectrum at zero.

References

- [1] J. Schober, *Regular one-parameter groups, reflection positivity and their application to Hankel operators and standard subspaces*, Ph.D. thesis, arXiv:2406.04241v2, 2025.